



ROBO LAB TECHNICAL REPORT SERIES

RL0020

Robot localisation in a rectangular room

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Problem statement

Given a mobile robot in a rectangular room of known size defined by the coordinates of its corners, we want the robot to locate itself within the room's coordinate system.

Specifically, given:

- A rectangular room is defined as a coordinate frame H with four corners that delimit the space.
- $N = 4$ fixed points $A_i \in \mathbb{R}^2$ defined in H , whose positions are initially given in a fixed coordinate frame $\mathbf{H}$.
- A robot is defined as a mobile point $R \in \mathbb{R}^2$ and with orientation $\phi \in \mathbf{SO}(2)$. This pose is defined in H . $\mathbf{SO}(2)$ is the space of all 2D rotations. It consists of all orthogonal matrices with a determinant of +1. The composed space of $\mathbf{R}^2$ and $\mathbf{SO}(2)$ is called $\mathbf{SE}(2)$
- N measurements m_i of the A_i points taken from the robot. Figure 1 shows graphically the problem elements. The corner positions are initially given and exact.

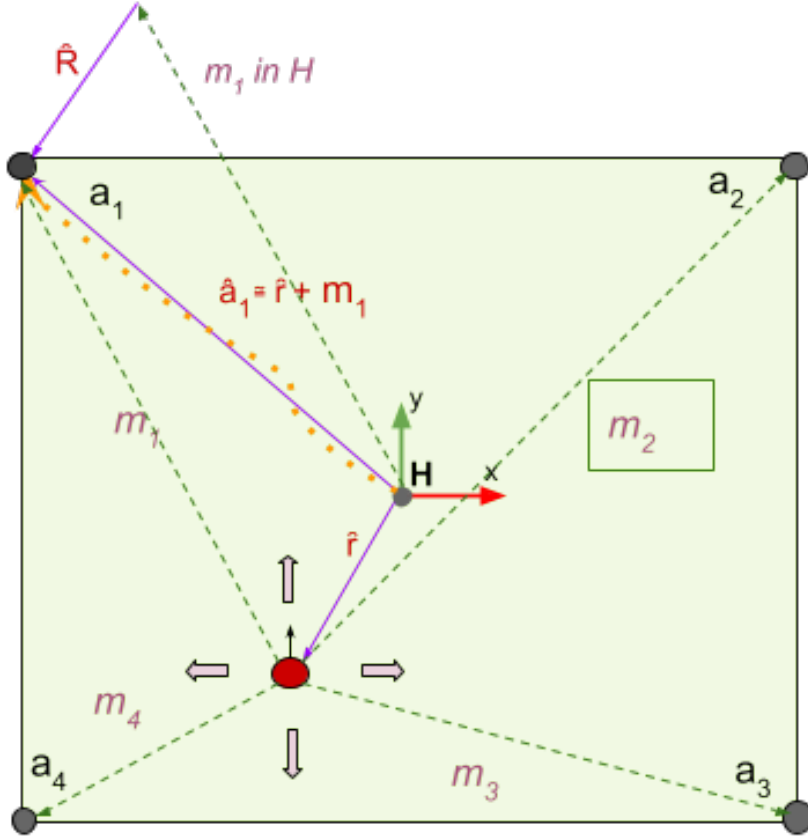


Figure 1: Problem configuration: four fixed points A_i are defined wrt the room centre H . The robot r is represented as a red circle in its estimated position. The four measurements m_i are drawn as curvy lines. We are interested in the error between the known vectors c_i and its estimation through the current measurement m_i taken at the unknown robot position $\hat{r}, c_i = \hat{c}_i + m_i$.

1 Objective

Estimate the full pose, translation and rotation of the robot M in the coordinate frame $\mathbf{H}$ by minimising an error function that considers the differences between each known (nominal) corner position and the corner position estimated from each measurement. We will have to match the measured corners to the nominal corners to

determine which corresponds to which. Measured corners come in the robot's coordinate frame, while nominal corners are given with respect to the room's coordinate frame. This implies that we have to transform one of the sets of corners into the other's coordinate frame.

2 Solution

We assume that measured corners match the nominal ones when the robot is correctly positioned. Since it moves and rotates, there will be a non-null transformation between the robot's frame and the room's frame. This can be expressed as,

$$c_i \approx \hat{c}_i = R_\phi \hat{m}_i + T_r \quad (1)$$

where

$$R_\phi = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix} \quad \text{and} \quad T_r = \begin{bmatrix} x \\ y \end{bmatrix} \quad (2)$$

The 2D matrix represents the robot's rotation, i.e., the angle it forms with the Y-axis of the room H , and T_r is the translation vector from H to the robot. Check that matrix R rotates any vector it multiplies on the right counterclockwise.

$y = Rx + T$ is the standard equation to represent an object x in the robot's frame, as seen by someone observing from H . To transform from H to the robot, we solve for $x = R^T(y - T)$. For rotation matrices, the inverse equals the transpose.

As you can see, this expression is non-linear on ϕ , since it is inside a $\sin()$ or $\cos()$ function. Assuming that the rotation error is small ($\pm 15^\circ$), we can linearise the angle as,

$$\cos\phi \approx 1 \quad \sin\phi \approx \phi \quad (3)$$

We get,

$$R(\phi) = \begin{bmatrix} 1 & -\phi \\ \phi & 1 \end{bmatrix} \quad (4)$$

Then,

$$\hat{c}_i = \begin{bmatrix} 1 & -\theta \\ \theta & 1 \end{bmatrix} \begin{bmatrix} m_{i,x} \\ m_{i,y} \end{bmatrix} + T_r \quad (5)$$

and developing the matrix multiplication,

$$\begin{bmatrix} c_{i,x} \\ c_{i,y} \end{bmatrix} = \begin{bmatrix} m_{i,x} - \theta m_{i,y} \\ \theta m_{i,x} + m_{i,y} \end{bmatrix} + \begin{bmatrix} r_x \\ r_y \end{bmatrix} \quad (6)$$

Gives,

$$\begin{aligned} c_{i,x} &= r_x + m_{i,x} - \theta m_{i,y} \\ c_{i,y} &= r_y + \theta m_{i,x} + m_{i,y} \end{aligned} \quad (7)$$

For multiple corners we get,

$$\underbrace{\begin{bmatrix} 1 & 0 & -m_{1,y} \\ 0 & 1 & m_{1,x} \\ 1 & 0 & -m_{2,y} \\ 0 & 1 & m_{2,x} \\ \vdots & \vdots & \vdots \end{bmatrix}}_W \underbrace{\begin{bmatrix} r_x \\ r_y \\ \theta \end{bmatrix}}_r = \underbrace{\begin{bmatrix} c_{1,x} - m_{1,x} \\ c_{1,y} - m_{1,y} \\ c_{2,x} - m_{2,x} \\ c_{2,y} - m_{2,y} \\ \vdots \end{bmatrix}}_b \quad (8)$$

which is a linear system of equations $Wr = b$ in the unknowns $[r_x, r_y, \phi]$. Since W is not square we cannot invert it to obtain $r = W^{-1}b$. We have to use the pseudoinverse ¹.

We first make W squared by left-multiplying by itself transposed,

$$W^T W r = W^T b \quad (9)$$

and now we can invert it,

$$\mathbf{r} = (W^T W)^{-1} W^T b \quad (10)$$

Since W is not diagonal, the solution $[r_x, r_y, \phi]$ now depends on the errors and the cross products involving ϕ , and ϕ depends on a weighted combination of cross-products between position errors and measured positions.

3 Implementation

We will implement Equation 10 using the Eigen math library ². Define a matrix W and a vector b as appears in Equation 8 and operate on them as indicated.

¹https://en.wikipedia.org/wiki/Moore-Penrose_inverse

²[https://en.wikipedia.org/wiki/Eigen_\(C++_library\)](https://en.wikipedia.org/wiki/Eigen_(C++_library))